

1. Find an inverse of a modulo m for each of these pairs of relatively prime integers.

1. $a = 4, m = 9$
2. $a = 19, m = 141$
3. $a = 55, m = 89$
4. $a = 89, m = 232$

Solution:

1. 7
2. 52
3. 34
4. 73

2. Show that if m is an integer greater than 1 and $ac \equiv bc \pmod{m}$, then $a \equiv b \pmod{m/\gcd(c, m)}$.

Solution: Let $d = \gcd(c, m)$, $m' = m/d$ and $c' = c/d$. Because all the common factors of m and c are divided out to obtain m' and c' , it follows that m' and c' are relatively prime. Because $m = m'd$ divides $ac - bc = (a - b)c = (a - b)c'd$, it follows that m' divides $(a - b)c'$. Since m' and c' are relatively prime, we see that m' divides $a - b$, so $a \equiv b \pmod{m'}$.

3. Use Fermat's little theorem to find $7^{121} \pmod{13}$.

Solution: Fermat's little theorem implies here that $7^{13-1} = 7^{12} \equiv 1 \pmod{13}$. Thus $(7^{12})^{10} \equiv 1^{10} \pmod{13}$ so $7^{120} \equiv 1 \pmod{13}$. Multiplying both sides by 7 gives the final answer of $7^{121} \equiv 7 \pmod{13}$, where $0 \leq 7 < 13$.

4. Use Fermat's little theorem to show that if p is prime and $p \nmid a$, then a^{p-2} is an inverse of a modulo p .

Solution: Fermat's little theorem implies here that $a^{p-1} \equiv 1 \pmod{p}$. Writing this as $a^{p-2} \cdot a \equiv 1 \pmod{p}$ implies the desired conclusion.

5. Find the greatest common divisor (gcd) and least common multiple (LCM) of the following pairs of numbers using their prime factorizations:

1. 72 and 120
2. 168 and 180

Solution: 72 and 120 $72 = 2^3 \times 3^2$,
 $120 = 2^3 \times 3 \times 5$. **gcd:** Take the lowest powers of common primes:

$$\gcd(72, 120) = 2^3 \times 3 = 24.$$

LCM: Take the highest powers of all primes:

$$LCM(72, 120) = 2^3 \times 3^2 \times 5 = 360.$$

168 and 180 $168 = 2^3 \times 3 \times 7$,
 $180 = 2^2 \times 3^2 \times 5$. **gcd:** Take the lowest powers of common primes:

$$\gcd(168, 180) = 2^2 \times 3 = 12.$$

LCM: Take the highest powers of all primes:

$$LCM(168, 180) = 2^3 \times 3^2 \times 5 \times 7 = 2520.$$

6. Find some composite number n such that there exists an integer a such that Fermat's Little Theorem does not hold for a in the sense that, there exists an a such that $\gcd(a, n) = 1$ and

$$a^{n-1} \not\equiv 1 \pmod{n}.$$

Solution: Let us take $n = 6$ and $a = 2$. Now, we compute $2^5 \pmod{6}$:

$$2^5 \equiv 32 \equiv 2 \not\equiv 1 \pmod{6}.$$